Bike Sharing Demand Prediction

Machine Learning Competition — Team ABA

MSc Statistics, University of Geneva — Fall 2024

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Abstract

A Geneva bike-sharing company (Mule) requires hourly rental forecasts for the next 3 months to support a bank loan application. Using a dataset from a comparable operator, we benchmark seven machine learning models — from linear regression to deep neural networks — on temporal, meteorological, and calendar features. The best model (Deep Neural Network, Kaggle MAE: **35.75**) is selected for final predictions. Feature importance analyses consistently identify **hour of day**, **temperature**, and **working day status** as the dominant demand drivers.

1. Introduction

Mule provides Geneva residents with hourly bicycle rentals. To secure bank financing, the company requires reliable sales forecasts for the following three months. Mule supplied a dataset from a similar operator in another country, containing temporal, meteorological, and calendar features alongside hourly rental counts. Reliable weather forecasts for the prediction window are also available.

We evaluate seven models of increasing complexity — Linear Regression, Lasso, Decision Tree, Random Forest, Gradient Boosting, Support Vector Regression, and a Deep Neural Network — benchmarked by **Mean Absolute Error (MAE)**. Beyond predictive performance, the analysis delivers actionable business insights by identifying the key drivers of rental demand.

2. Data & Preprocessing

2.1 Dataset Structure

The training set contains **15,211 rows** and **15 columns**, covering hourly observations over two years. After imputation, one row is dropped (first observation, no prior humidity value), leaving **15,210 rows**. The test set contains no missing values.

Feature	Type	Description
season	Categorical	Fall (0), Winter (1), Spring (2), Summer (3)
yr	Binary	2011 (0), 2012 (1)
mnth / hr / weekday	Cyclic	Encoded as sine/cosine transforms (see §2.3)
holiday / workingday	Binary	No (0) / Yes (1)
weathersit	Ordinal	Clear (0) → Heavy rain/snow (3)
temp / hum / windspeed	Numerical	Normalised in [0, 1]
cnt	Target	Total hourly rentals (casual + registered)

Table 1 — Feature overview after encoding.

2.2 Missing Value Imputation

Missing values are present in all columns except Id and cnt. Imputation strategy:

- **dteday:** inferred from adjacent rows using the hour column.
- **season, yr, mnth, holiday, weekday, weathersit:** forward-fill after verifying no period boundaries are crossed.
- **hr:** incremented from the previous observation; resets to 0 after 23.
- **workingday:** derived from the weekday column.
- **temp, atemp, hum, windspeed:** interpolated as the average of the nearest non-NA neighbours.

2.3 Feature Engineering

Dropped: `Id` and `dteday` (redundant with other temporal columns); `atemp` (Pearson r = 0.99 with `temp` — removes multicollinearity in linear models and redundancy in tree-based models).

Cyclic encoding: `hr`, `weekday`, and `mnth` are circular by nature. Sine and cosine transforms preserve this structure:

```
hour_sin = sin(2π · hr / 24),    hour_cos = cos(2π · hr / 24)
```

The same transformation is applied to weekday (period 7) and mnth (period 12).

3. Exploratory Data Analysis

3.1 Numerical Feature Distributions

Temperature is slightly left-skewed — observations cluster at higher values, reflecting more data in warmer months. Humidity follows a roughly uniform distribution. Windspeed is right-skewed, confirming that extreme wind events are rare. The rental count `cnt` is heavily right-skewed: high demand spikes are uncommon.

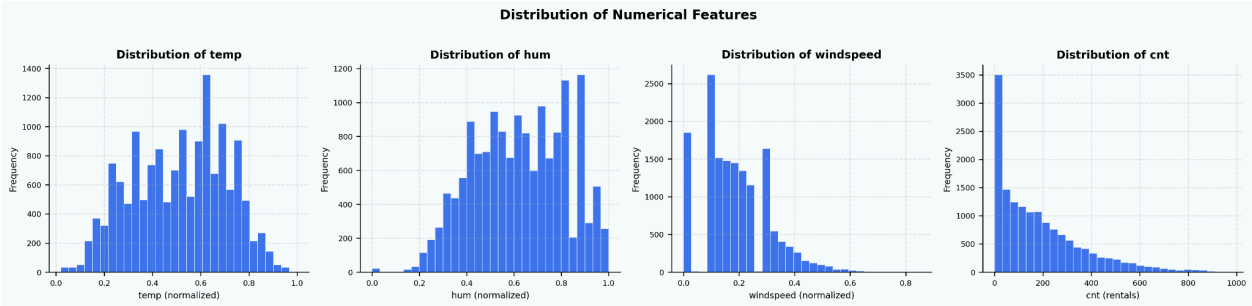

Figure 1 — Distribution of numerical features. Weather variables are normalised to [0, 1].

3.2 Hourly Rental Patterns

Figure 2 reveals two sharp demand peaks on working days: **7:00–9:00** and **16:00–18:00**, corresponding to morning and evening commutes. Summer consistently yields the highest counts; Winter the lowest, reflecting the strong positive correlation between temperature and demand.

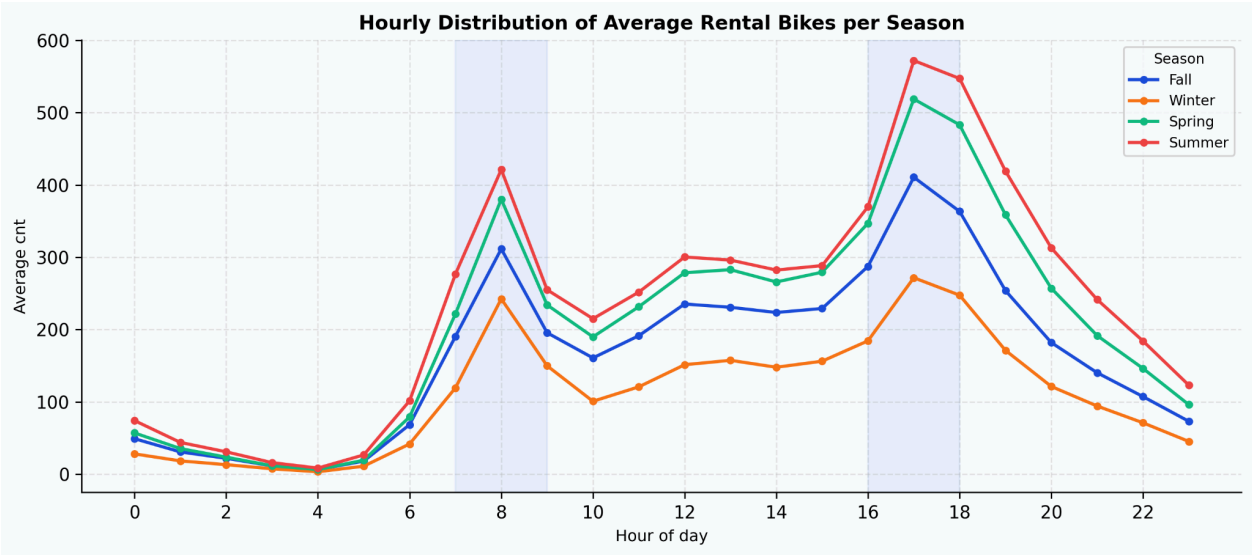

Figure 2 — Hourly distribution of average rental bikes per season. Shaded areas indicate rush hours.

Figure 3 contrasts weekdays with weekends and holidays. On workdays the bimodal commute pattern is clear; on weekends demand builds gradually through the afternoon with no sharp peaks. This distinction motivates retaining `workingday` as a feature.

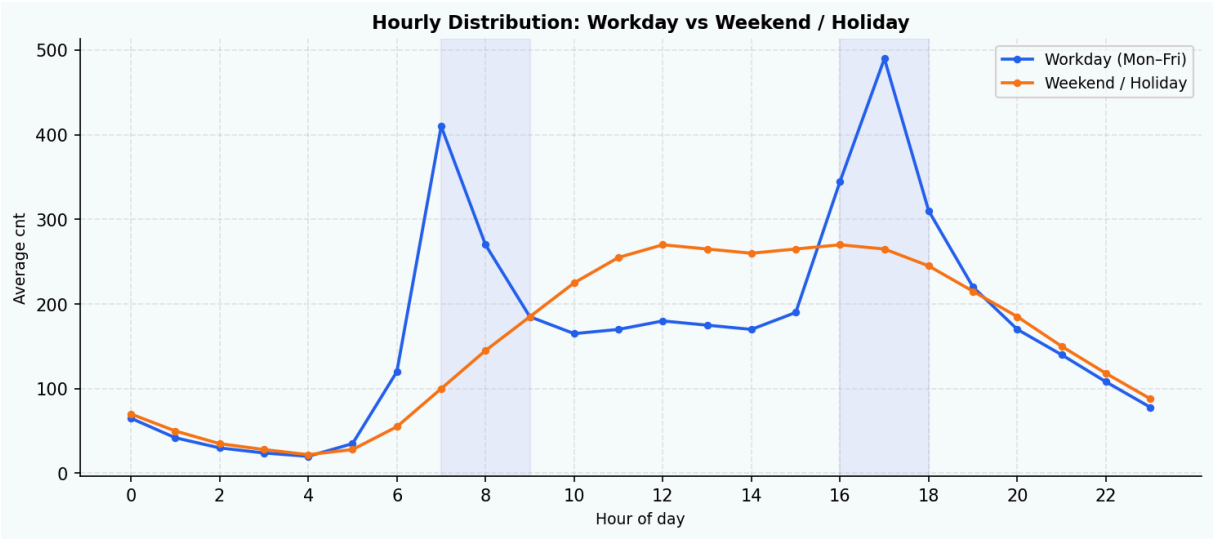

Figure 3 — Hourly distribution: workday vs weekend / holiday.

3.3 Monthly Trend and Year-over-Year Growth

Average rentals peak between **June and September** (months 5–8), driven by warmer temperatures and longer daylight. The violin plot (Figure 5) shows that 2012 has a substantially wider distribution at higher count values, indicating significant year-over-year demand growth — a positive signal for Mule.

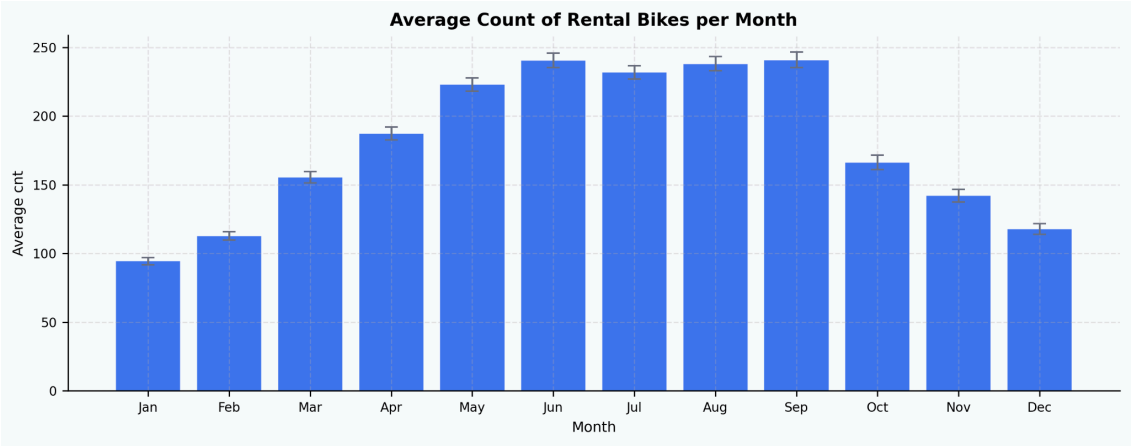

Figure 4 — Average rental count per month.

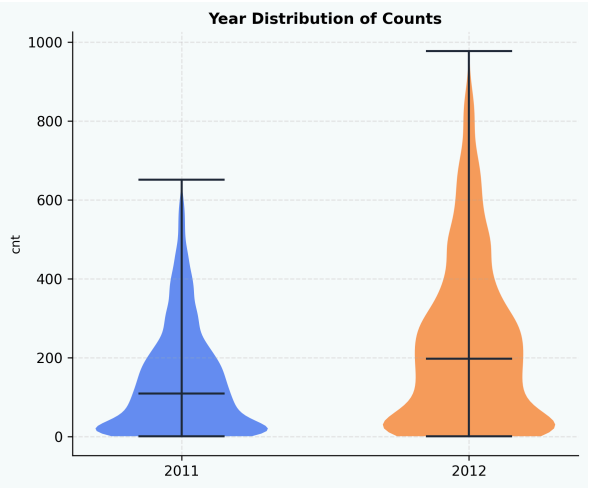

Figure 5 — Count distribution by year (violin plot).

3.4 Correlation Analysis

The rental count is most strongly correlated with **temp** (r = 0.44), **hr cyclic features** (r ≈ 0.40), and **yr** (r = 0.27). Negative correlations: **hum** (r = −0.33) and **weathersit** (r = −0.15). The near-perfect collinearity between temp and atemp (r = 0.99) confirms the decision to drop atemp.

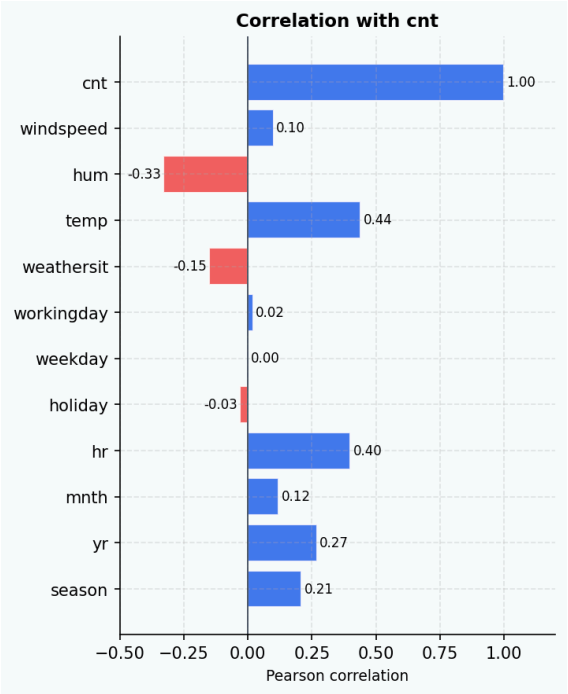

Figure 6 — Pearson correlation of each feature with the target cnt.

4. Modelling

All models are evaluated on an 80/20 train/test split (random state = 42). Hyperparameter selection uses 5-fold cross-validation. The **1-SE rule** is applied where relevant: among models within one standard error of the best CV score, the simplest is preferred to reduce overfitting risk.

4.1 Linear Regression

Linear regression serves as the interpretable baseline. It achieves a test MAE of **90.57**, confirming that the relationship between features and rental counts is strongly non-linear. Feature coefficients identify `temp`, `yr`, and `workingday` as the most influential predictors, consistent with EDA findings.

4.2 Lasso Regression

Lasso incorporates an L1 penalty that shrinks irrelevant coefficients to exactly zero, providing automatic feature selection. We choose Lasso over Ridge because both yield similar predictive performance while Lasso produces sparser, more interpretable solutions. Alpha is selected via grid search with the 1-SE rule.

Test MAE: **90.55**. Features zeroed out by Lasso: `workingday`, `windspeed`, `weekday_sin` — these contribute minimally beyond what retained features already capture.

4.3 Decision Tree (depth = 5)

A single tree constrained to `max_depth=5` balances interpretability and generalisation. Test MAE: **63.77**. Figure 7 shows that `hour_cos` and `hour_sin` together account for over 65% of feature importance, underscoring the primacy of time-of-day in demand prediction.

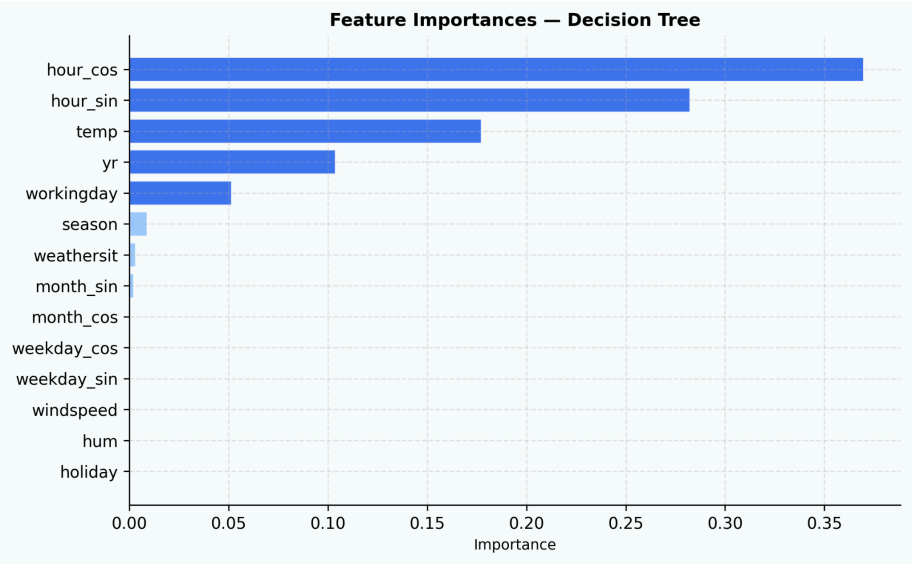

Figure 7 — Feature importances for the Decision Tree (max depth = 5).

4.4 Random Forest

Random Forest mitigates single-tree overfitting by averaging predictions from multiple decorrelated trees, each trained on a bootstrap sample. At each split, only a random subset of m features is considered (heuristic: $m \approx p/3$).

Grid search over n_estimators $\in$ {100, 200, 300}, max_depth $\in$ {12, 15, 18}, max_features $\in$ {4, 5, 6}. Best configuration: **300 estimators, depth 18, 5 features**. Test MAE: **25.49** | Kaggle MAE: **54.37**.

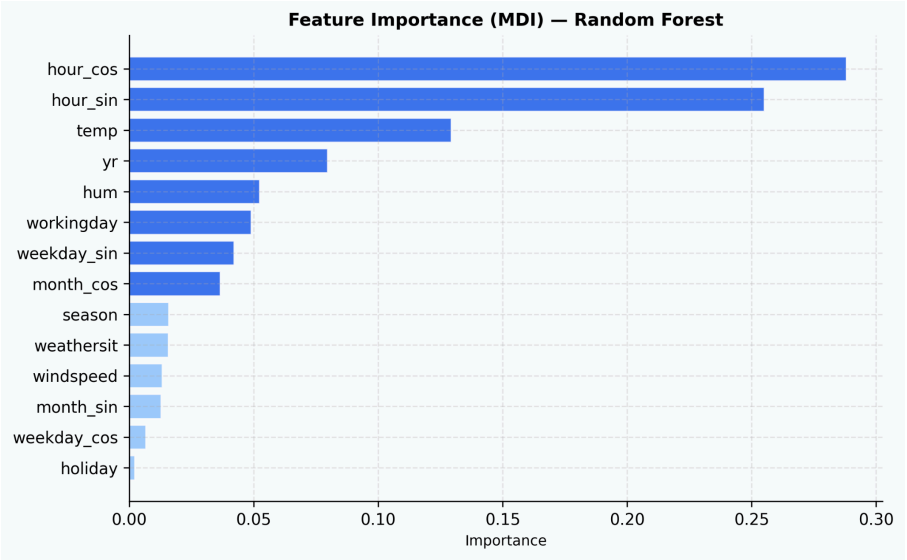

Figure 8 — Feature importance (MDI) for the Random Forest.

4.5 Gradient Boosting

Gradient Boosting builds trees sequentially, each correcting the residuals of its predecessor. This iterative bias reduction makes it more expressive than Random Forest, but its sequential nature increases overfitting risk.

Two-stage grid search over n_estimators, learning_rate, and max_depth with the 1-SE rule. Best result: **1000 estimators, lr = 0.1, depth 5**. Test MAE: **24.44** | Kaggle MAE: **43.32**. The gap between local and leaderboard MAE suggests mild overfitting to the test split.

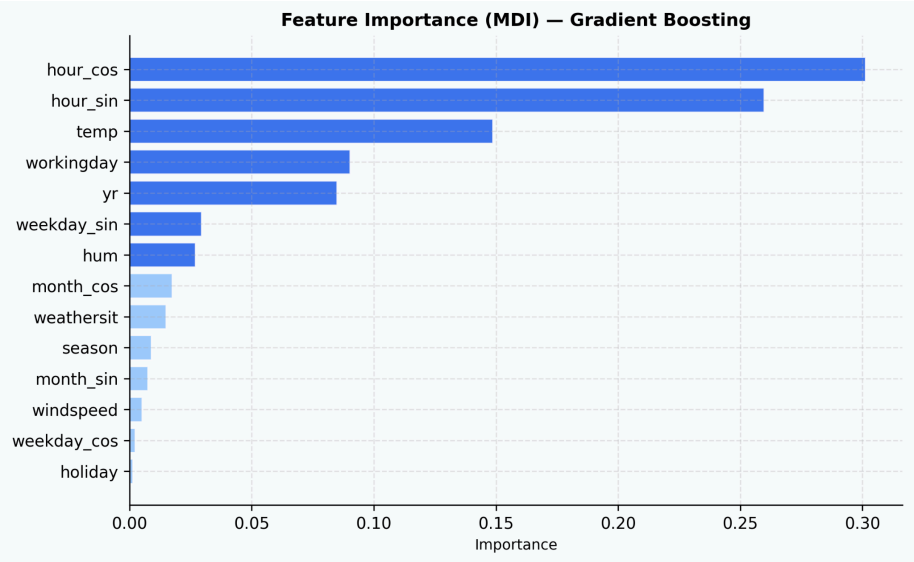

Figure 9 — Feature importance (MDI) for Gradient Boosting.

4.6 Support Vector Regression (SVR)

SVR with an RBF kernel captures non-linear interactions via a higher-dimensional feature space. The ε-insensitive loss makes the model robust to noise. All features are standardised prior to fitting.

Best parameters: **C = 700, ε = 0.7**. Test MAE: **43.41** | Kaggle MAE: **65.85**. SVR underperforms ensemble methods and incurs high computational cost; it was not optimised further.

4.7 Deep Neural Network

Architecture selected via 5-fold CV comparing 2, 3, and 4 hidden-layer configurations. The 3-layer model (256 → 128 → 64 neurons) achieves comparable validation MAE to the 4-layer model with substantially lower variance, and is preferred on parsimony grounds.

Layer	Units	Activation	Regularisation
Input	19	—	—
Hidden 1	256	LeakyReLU (α=0.1)	BatchNorm + Dropout(0.2)
Hidden 2	128	LeakyReLU (α=0.1)	BatchNorm + Dropout(0.2)
Hidden 3	64	LeakyReLU (α=0.1)	—
Output	1	Linear	—

Table 2 — DNN architecture. Optimizer: Adam. Loss: MSE. Callbacks: EarlyStopping + ReduceLROnPlateau.

LeakyReLU prevents the dying neuron problem by allowing small non-zero gradients for negative inputs. **Batch Normalisation** stabilises training by normalising layer activations. **Dropout** (rate = 0.2) prevents co-adaptation, acting as an implicit ensemble regulariser.

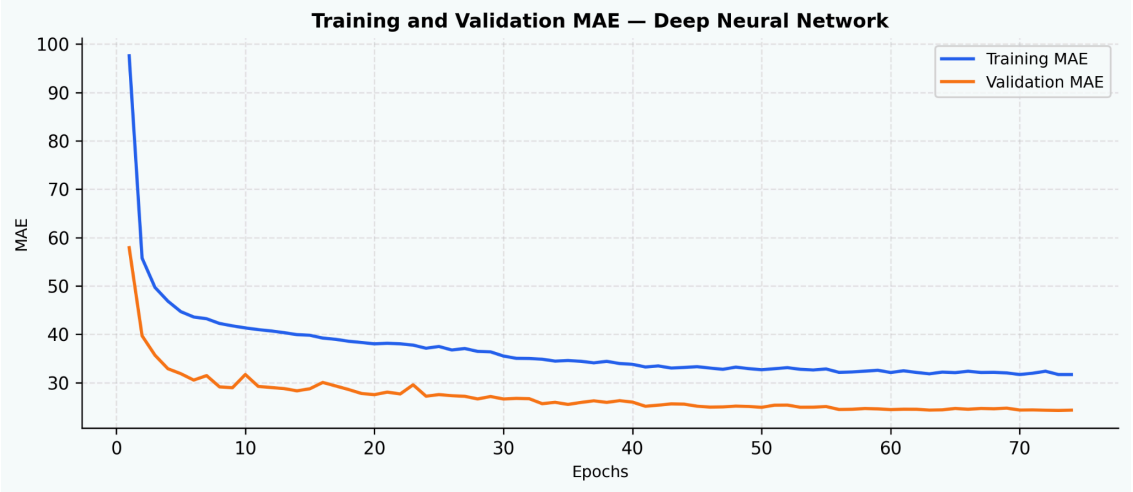

Figure 10 — Training and Validation MAE over epochs. Early stopping prevents overfitting.

Final Kaggle MAE: **35.75** — best result across all models.

5. Results & Conclusion

5.1 Model Comparison

Model	Test MAE	Kaggle MAE	Notes
Linear Regression	90.57	—	Baseline; linear only
Lasso Regression	90.55	—	Feature selection via L1
Decision Tree (depth=5)	63.77	—	Interpretable; constrained depth
Random Forest	25.49	54.37	300 trees, depth 18, 5 features
Gradient Boosting	24.44	43.32	1000 est., lr=0.1, depth 5
SVR (RBF kernel)	43.41	65.85	C=700, ε=0.7; high cost
Deep Neural Network	**24.09**	**35.75**	**256→128→64, LeakyReLU, Dropout**

Table 3 — Model performance. Best Kaggle MAE (DNN) highlighted in blue.

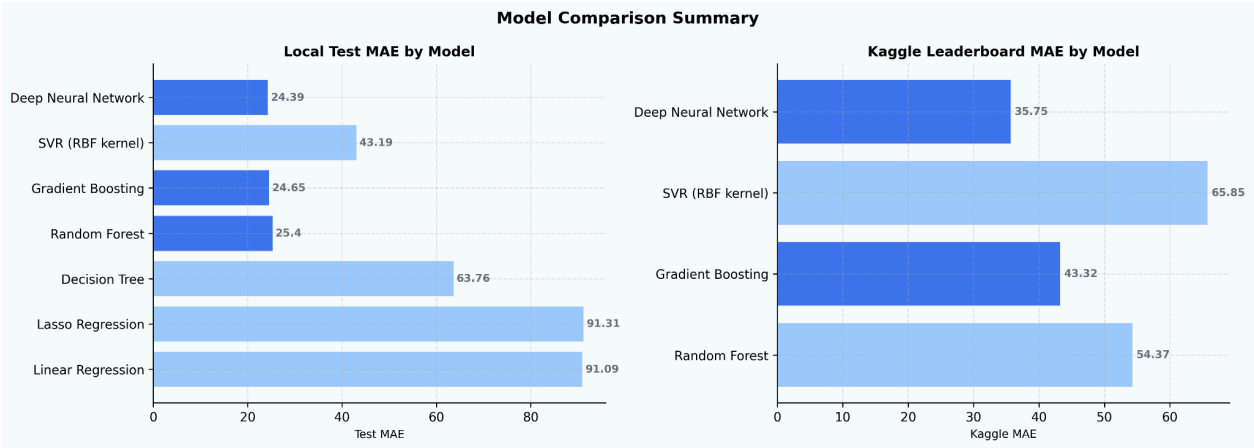

Figure 11 — Local test MAE (left) and Kaggle leaderboard MAE (right) by model.

5.2 Key Business Insights for Mule

Feature importance analyses across Decision Tree, Random Forest, and Gradient Boosting consistently point to three dominant demand drivers:

- **Time of day** (hour_cos, hour_sin): strongest predictor. Demand peaks at 7:00–9:00 and 16:00–18:00 on working days; a smooth afternoon pattern on weekends. Mule should ensure maximum fleet availability during commute windows.
- **Temperature**: strong positive correlation (r = 0.44). Seasonal inventory planning should account for significant demand reduction in winter.
- **Working day status**: weekday and weekend usage patterns are fundamentally different, requiring distinct operational strategies for fleet distribution and rebalancing.

5.3 Forecast Outlook

The DNN model was applied to the test set covering Fall and Winter 2012. Despite lower seasonal temperatures, predicted counts are substantially higher than the equivalent period in 2011, reflecting the strong year-over-year growth trend observed in training data. Mule can expect continued demand growth and should plan fleet expansion accordingly.

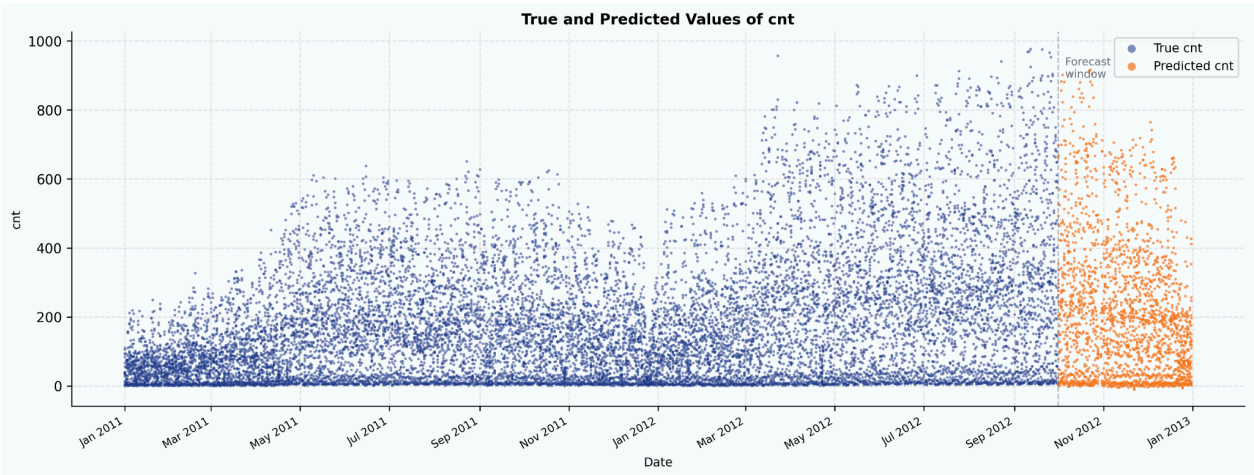

Figure 12 — DNN Forecast of Rental bikes for Winter 2012

6. References

- Kaggle competition: ML-UNIGE Fall 2024 — Bike Sharing.
  https://www.kaggle.com/competitions/ml-unige-fall-2024-bike-sharing
- Machine Learning course (S403011), MSc Statistics, University of Geneva — Fall 2024.